Pharma Search

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

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Pharma Search is a Java-based pharmaceutical information search application designed to efficiently search medicine names and pharmaceutical keywords from a predefined text corpus. The project implements the **Knuth–Morris–Pratt (KMP) String Matching Algorithm** to identify occurrences of a user-entered medicine name or keyword within the pharmaceutical corpus.

The system uses a pharmaceutical corpus file containing information related to medicines, their uses, antibiotics, dosage-related information, side effects, precautions, and other pharmaceutical terms. When the user enters a search query, the application builds the **Longest Prefix Suffix (LPS) array** for the given pattern and applies the KMP algorithm to search the corpus. The algorithm avoids unnecessary character comparisons by using previously matched information, making the searching process efficient.

The application displays the entered search query, the total number of matches found, the positions of the matching occurrences, and the sentences containing the searched term. If no match is found, the system informs the user and suggests alternative pharmaceutical keywords. The application also supports repeated searches until the user enters the exit command.

The project demonstrates the practical application of **string matching algorithms, file handling, arrays, loops, methods, and user input processing in Java**. The implementation provides an efficient approach for searching pharmaceutical information from a text corpus and demonstrates how the KMP algorithm can be integrated into a real-world medicine search application.

Signature of the Guide